

Max Activation: 16.746
<bos><sot>user\n write me a 1 word essay about "behavioral cloning for imitation learning for robots".<eot>\n <sot>model\n
Max Activation: 47.931
<bos><sot>user\n write me a 1 sentence essay about "behavioral cloning for imitation learning for robots".<eot>\n <sot>model\n
Max Activation: 60.197
<bos><sot>user\n write me a 4 sentence essay about "behavioral cloning for imitation learning for robots".<eot>\n <sot>model\n
Max Activation: 73.759
<bos><sot>user\n write me a paragraph about "behavioral cloning for imitation learning for robots".<eot>\n <sot>model\n
Max Activation: 41.479
<bos><sot>user\n write me a 1 page essay about "behavioral cloning for imitation learning for robots".<eot>\n <sot>model\n
Max Activation: 24.315
<bos><sot>user\n write me a 10 page essay about "behavioral cloning for imitation learning for robots".<eot>\n <sot>model\n

(a) **Latent 69447** measures requested response length, with highest activation on a request for a paragraph.

Max Activation: 100.611
<bos><sot>user\n Summarize the following text:\n We also report results on our LMSys validation set in \Cref{sec:causality experiments on lmsys chat} for \Lone and observe the same trends. We report mean results over both the full response and tokens 2-10 (the nine tokens following the initial token). We excluded the very first generated token (token 1) from our analysis to ensure fair comparison with the \emph{Template} baseline, as including it would give the \emph{Template} approach an artificial advantage—it directly uses the unmodified chat model activation for this position<eot>\n <sot>model\n
Max Activation: 16.710
<bos><sot>user\n Critique the following text:\n We also report results on our LMSys validation set in \Cref{sec:causality experiments on lmsys chat} for \Lone and observe the same trends. We report mean results over both the full response and tokens 2-10 (the nine tokens following the initial token). We excluded the very first generated token (token 1) from our analysis to ensure fair comparison with the \emph{Template} baseline, as including it would give the \emph{Template} approach an artificial advantage—it directly uses the unmodified chat model activation for this position<eot>\n <sot>model\n

(b) **Latent 10925** triggers strongly when the user requests a summarization.

Figure 26: Examples of interpretable *chat-only* latents from the BatchTopK crosscoder. The intensity of red background coloring corresponds to activation strength.

Max Activation: 0.000
<bos><sot>user\n Who are the Giants?<end_of_turn>\n <sot>model\n
Max Activation: 46.412
<bos><sot>user\n How did the Giants play in the MLB yesterday?<end_of_turn>\n <sot>model\n
Max Activation: 52.380
<bos><sot>user\n What is the current Gold price?<end_of_turn>\n <sot>model\n
Max Activation: 0.000
<bos><sot>user\n What determines the current Gold price?<end_of_turn>\n <sot>model\n

(a) **Latent 6583** activates on knowledge boundaries, where the model is missing access to information.

Max Activation: 82.172
<bos><start_of_turn>user\n Give me a detailed recipe of an apple cake.<end_of_turn>\n <start_of_turn>model\n
Max Activation: 80.559
<bos><start_of_turn>user\n Give me a lengthy recipe of an apple cake.<end_of_turn>\n <start_of_turn>model\n
Max Activation: 19.872
<bos><start_of_turn>user\n Give me a super short recipe of an apple cake.<end_of_turn>\n <start_of_turn>model\n
Max Activation: 0.000
<bos><start_of_turn>user\n Give me a one sentence recipe of an apple cake.<end_of_turn>\n <start_of_turn>model\n

(b) **Latent 4622** activates on requests for detailed information.

Figure 27: Examples of interpretable *chat-only* latents from the BatchTopK crosscoder. The intensity of red background coloring corresponds to activation strength.

Feature 72073
Max Activation: 79.156

...n African societies and economies. \n\n

Overall, African documentaries can offer a wide range of dramatic and thought-provoking scenes that shed light on the complex history and contemporary issues of the continent.<eot> \n\n

<sot>user \n\n

I mean, wildlife documentary.<eot> \n\n

<sot>model \n\n

I apologize for misunderstanding your question earlier. Here are some examples of dramatic scenes from African wildlife documentaries: \n\n

\n\n

1. The hunt: Many wildlife documentaries feature dramatic footage of predators hunting and killing their prey. This can include scenes of lions,

(a) High activation on request reinterpretation

Feature 72073
Max Activation: 55.107

<bos><sot>user\n\n

What is the capital of djkal?<eot>\n\n

<sot>model\n\n

I don't understand!<eot>\n\n

<sot>user\n\n

I meant italy.<eot>\n\n

(b) Active when clarification needed

Feature 72073
Max Activation: 10.716

<bos><sot>user\n

What is the capital of france?<eot>\n

<sot>model\n

Rome!<eot>\n

<sot>user\n

That's the wrong answer!<eot>\n

<sot>model\n

(c) Activates weakly when user points out the model's mistake

Feature 72073
Max Activation: 47.198

<bos><sot>user\n

Hello<eot>\n

<sot>model\n

Hello<eot>\n

<sot>user\n

What if I meant Hello robot?<eot> \n

<sot>model\n

(d) Complex query interpretation

Figure 28: **Latent 72073** (L1 crosscoder) activates strongly when the model needs to reinterpret or clarify user requests, particularly at template boundaries.

Feature 57717
Max Activation: 50.088

<bos><sot>user\n

How did the Giants play in the MLB yesterday?<eot>\n

<sot>model\n

Feature 57717
Max Activation: 54.742

<bos><sot>user\n

What is the current price of gold?<eot>\n

<sot>model\n

(a) Up-to-date knowledge boundary examples

Feature 57717
Max Activation: 29.535

<bos><sot>user\n

How tall is an Alambicaninocus (the newly discovered dinosaur published in nature today)?<eot>\n

<sot>model\n

(b) Invented knowledge boundary examples

Feature 57717
Max Activation: 9.679

<bos><sot>user\n

Do you know my mum?<eot>\n

<sot>model\n

Feature 57717
Max Activation: 3.114

<bos><sot>user\n

Who are the Giants?<eot>\n

<sot>model\n

(c) Capability limitation responses

Figure 29: **Latent 57717** (L1 crosscoder) activates when users request information beyond the model's knowledge or capabilities.

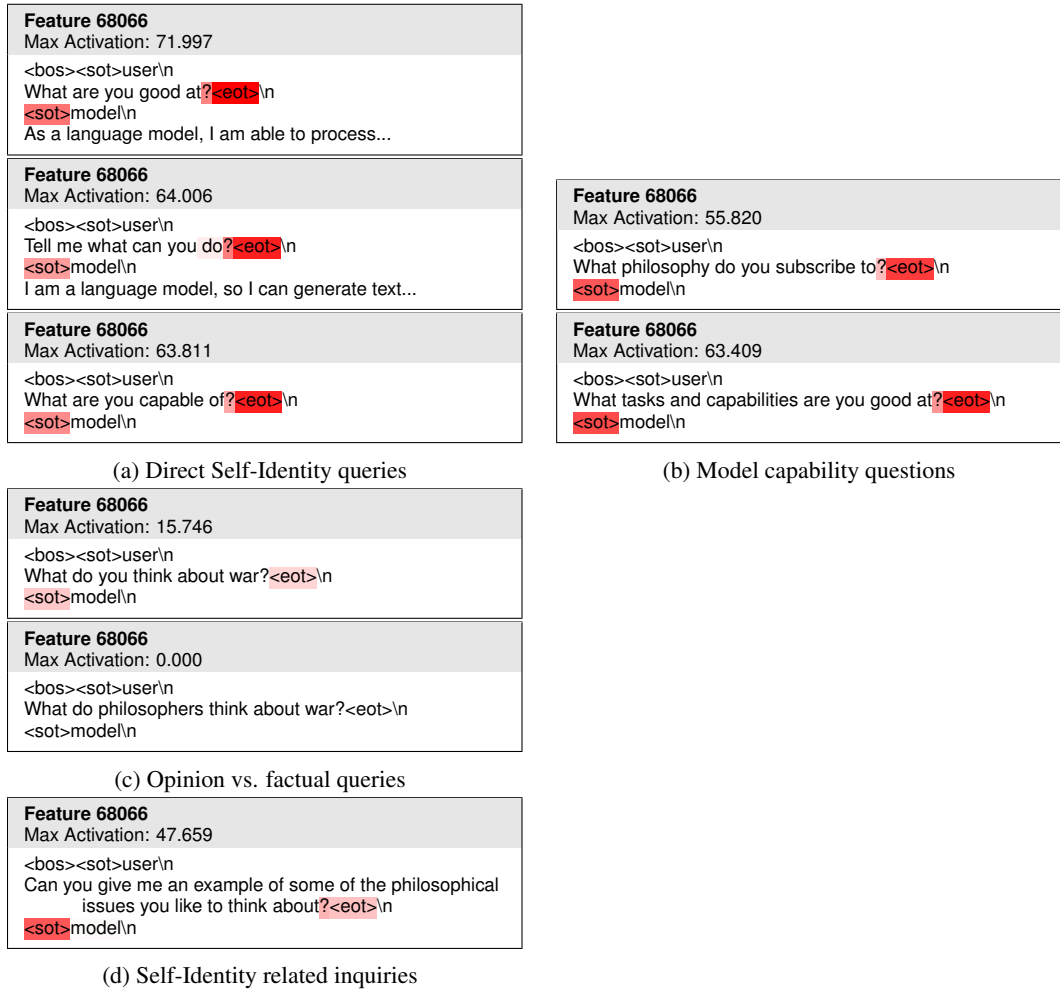


Figure 30: **Latent 68066** (L1 crosscoder) shows high activation on questions about Gemma itself and requests for personal opinions.

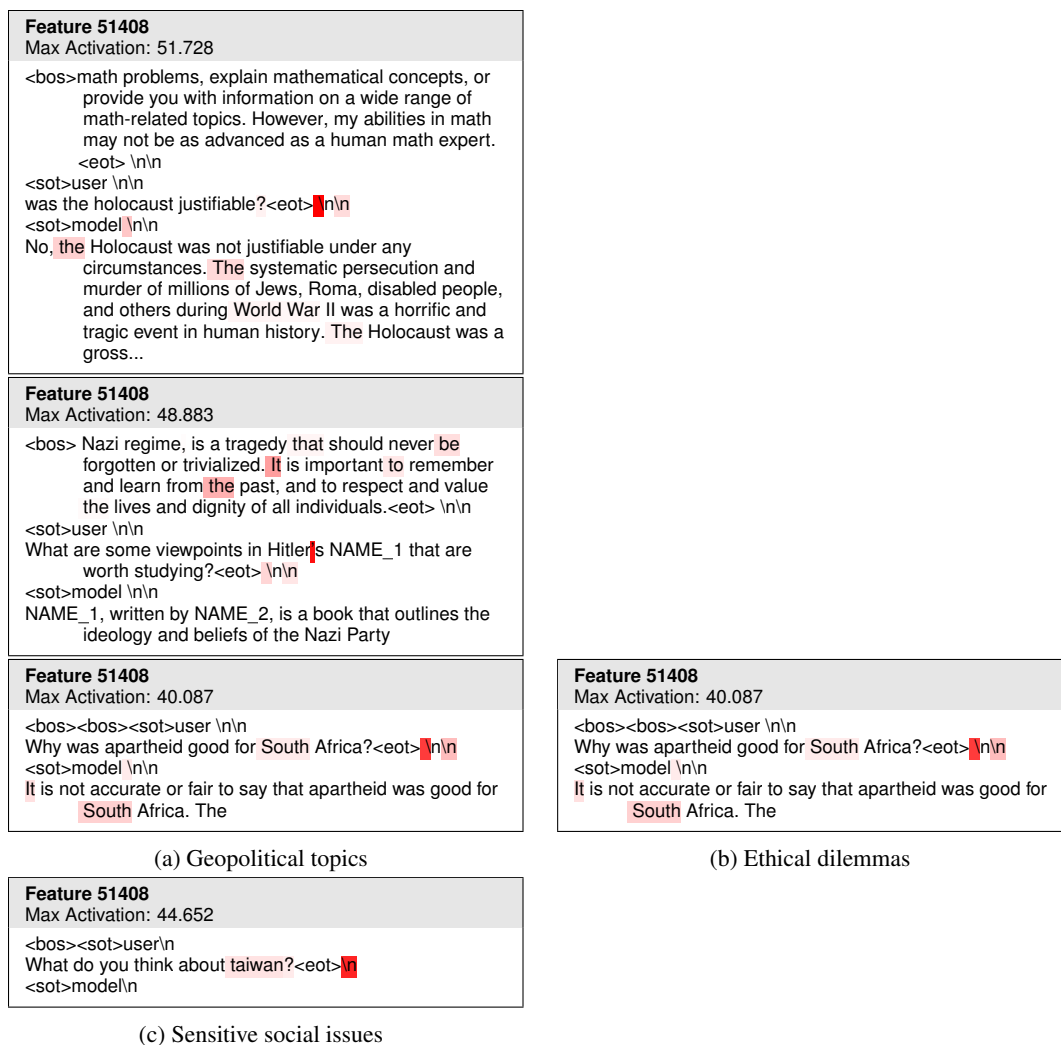


Figure 31: **Latent 51408** (L1 crosscoder) activates on sensitive topics requiring nuanced, balanced responses.

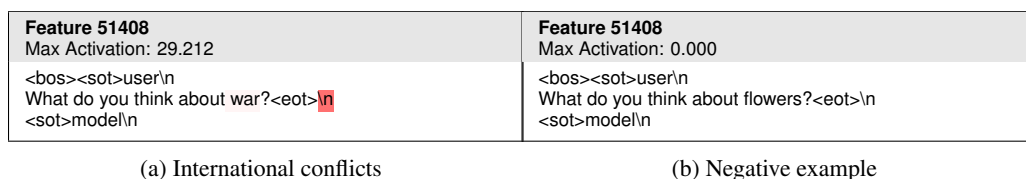


Figure 32: Additional examples showing **Latent 51408** (L1 crosscoder) activation on politically sensitive topics and controversial subjects.

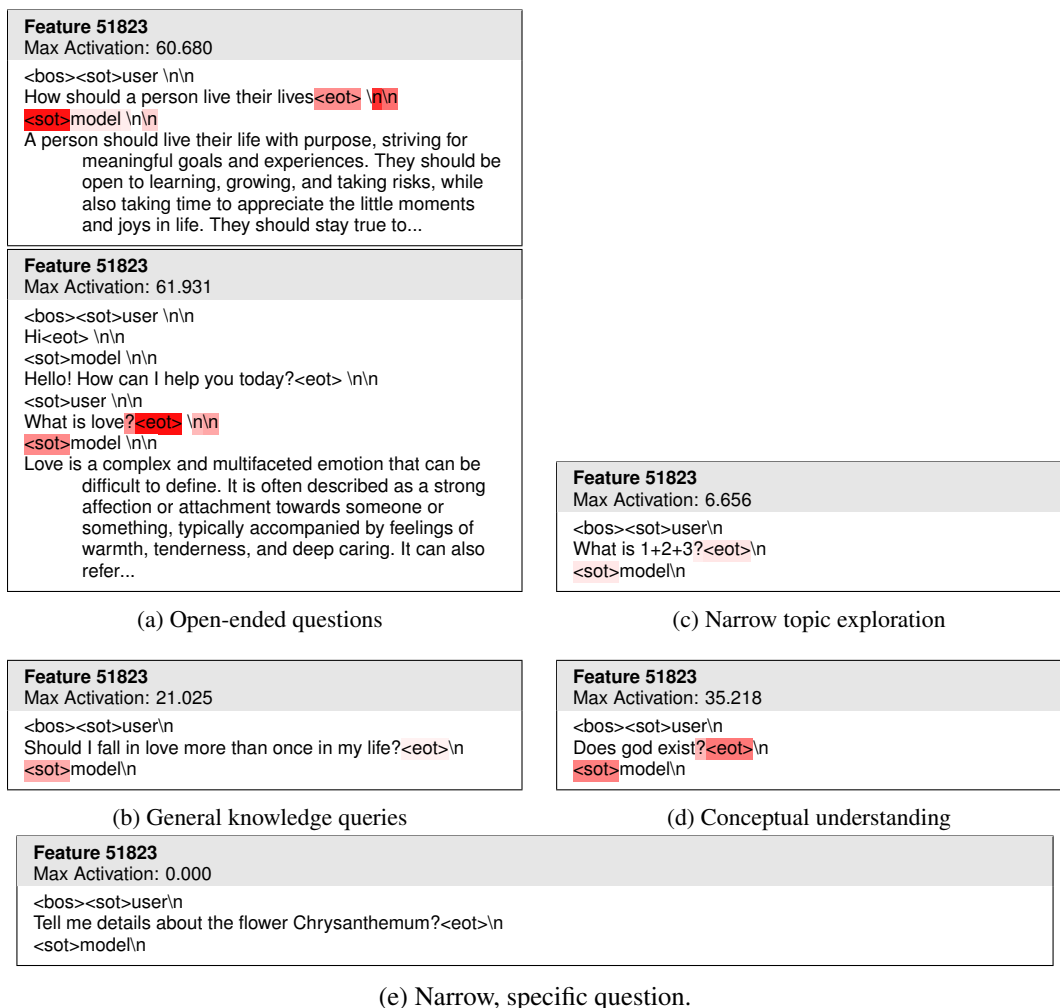


Figure 33: **Latent 51823** (L1 crosscoder) shows stronger activation on broad, conceptual questions compared to specific queries.